LANDCROS



HCM COUNTERMEASURE ROLL-OUT 2027

APP USER MANUAL

A step-by-step guide for supervisors, admins and viewers of the countermeasure tracker for the FQM fleet (EH4000AC-3 · EH3500AC-2 · EX5600-7). Written for first-time users, with worked examples and troubleshooting tips.

VERSION	**1.1**
ISSUED	**13 November 2026**
LANGUAGE OF THE APP	**English / Japanese**

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1. WELCOME

WHAT THE APP DOES

The **HCM Countermeasure Roll-out 2027** app tracks how each of the HCM countermeasures (weld reinforcements, brackets, seat screws, pedals...) is being applied across the FQM fleet: about 100 machines of the models EH4000AC-3, EH3500AC-2 and EX5600-7.

It replaces the Excel *Countermeasures Progress Tracker* and the *Action & Roll-out Plan*. Everybody works on the same live data, every change is recorded with who did it and when, and the dashboards, Gantt charts, alerts and reports update automatically.

WHO USES IT

ROLE	WHAT THEY DO	HOW THEY GET IN
Viewer	Sees the read-only landing page: KPIs, timelines, truck view, action plan, Top page. Can download the action plan PDF.	Just opens the app link. No account needed.
Supervisor	Records the field progress of every truck: planned, execution steps (e.g. RHS / LHS), FIR uploaded, report / timesheets, dates, work order, notes.	Signs in with a username and password.
Admin	Everything a supervisor does, plus: summary and alerts, edits the action plan, adds or removes countermeasures and trucks, marks N/A, sets working hours, publishes reports, settings.	Signs in with an admin account.
Master (app owner)	Can switch between the Admin and Supervisor views. Creates the user accounts.	Account defined in the app secrets.

> **TIP**
> If this is your first time, read chapters 2 and 3 first. Supervisors then go to chapter 5, admins to chapters 5 and 6. Chapter 9 explains what to do when something does not update.

KEY WORDS USED IN THIS MANUAL

WORD	MEANING
Countermeasure	A modification that must be applied to the trucks, e.g. Front Nose Cone Additional Weld. Each has a responsible person, a fleet target date and a duration per machine.
Truck record (truck-job)	One truck x one countermeasure. RD169 + Front Nose Cone is one record.
Execution steps	The parts of the work that are ticked when done. Most countermeasures have one step (Executed); welds often have two (e.g. RHS Front and LHS Front).
Planned	Tick that shows the job is scheduled. It is ticked automatically when you enter a planned date or tick all execution steps.
FIR uploaded / Report	The two documentation steps. A job is only Closed out when both are ticked.
N/A	The countermeasure does not apply to that truck. It is excluded from every %. Only admins set it.
Fleet target	Date by which every applicable truck should have the countermeasure executed.
Working hours	Estimated hours of work per truck for a countermeasure (same as Duration per machine). Used for remaining work, downtime and completion estimates.

2. GETTING STARTED

OPENING THE APP

1. Open the link shared by the app owner in any modern browser (Chrome, Edge, Safari) on a PC, tablet or phone. The online app runs on Streamlit Community Cloud.
2. The first page you see is the **public landing page** (chapter 4). It is read-only: you can look at everything without signing in.
3. If the app has not been used for a while it may show *"This app has gone to sleep"*. Click **Yes, get this app back up!** and wait about 30-60 seconds (see chapter 9).

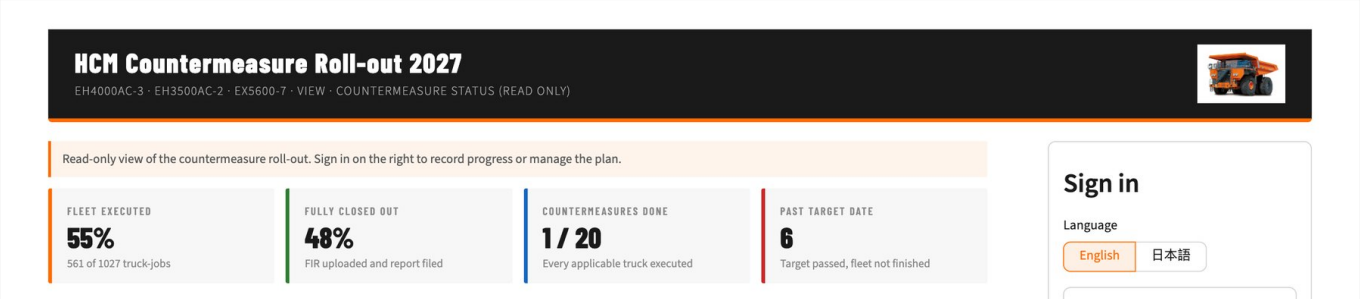


Figure 1. Top of the landing page: LANDCROS header, fleet KPI cards and the sign-in panel on the right.

SIGNING IN

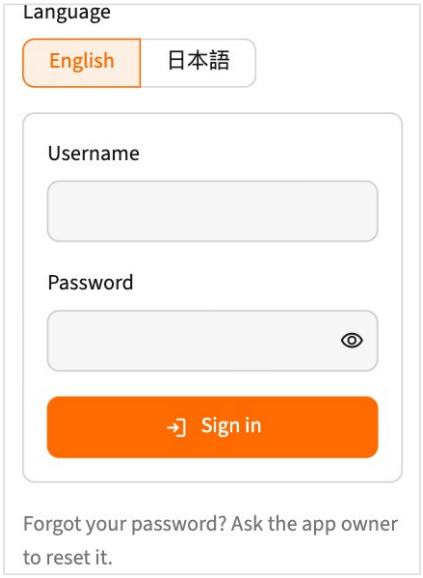


Figure 2. Sign-in panel. Choose the language, type your username and password, press Sign in.

1. On the right side of the landing page, choose **English** or **Japanese**.
2. Type your **Username** (e.g. *marcos1*; capital letters and spaces are ignored, so *Marcos1* or *Peter 1* also work) and your **Password** (capital letters matter).
3. Press **Sign in** (or press Enter). You land on your own view: supervisors on the progress editor, admins on the Summary tab.

> NOTE
> After a wrong password the app waits a few seconds before you can try again (1 s, 2 s, up to 5 s). This protects the accounts. If you forgot your password, ask the app owner to reset it - nobody, not even the owner, can see your current password.

THE SIDEBAR



Figure 3. Sidebar after signing in: your name and role, language, Refresh data, Change my password, Sign out, where the data is stored and today's date.

ITEM	WHAT IT DOES
Name and role	Shows who is signed in (Supervisor, Admin or Master).
Language	Switches every screen between English and Japanese. Your data does not change.
View (master only)	Switches between the Admin and Supervisor views.
Refresh data	Reloads the latest data saved by all users. Use it whenever you expect to see somebody else's change.
Change my password	Supervisors change their password here. Admins do it in the Settings tab.
Sign out	Ends your session. Always sign out on shared computers.

Data / Today	Where the data is stored (GitHub for the live app) and today's date in the project time zone (Africa/Lusaka by default).

TIP

If the sidebar is hidden (small screens), click the orange **>>** button at the top-left corner to open it again.

3. HOW PROGRESS IS MEASURED

STATUS OF A TRUCK RECORD

Every truck record gets a status from the boxes ticked. The same colours are used in the Top page matrix, the tables and the reports.

STATUS	WHEN	SYMBOL IN TOP PAGE
Closed out	All execution steps + FIR uploaded + Report / timesheets ticked.	tick
Executed - docs open	All execution steps ticked, but FIR or report still missing.	E
In progress	Something ticked (Planned, a step, FIR or report) but not all execution steps.	half circle
Not started	Nothing ticked.	dot
N/A	Countermeasure does not apply to this truck. Not counted anywhere.	N/A

PROGRESS, FLEET % AND COUNTERMEASURE STATUS

- **Progress of a record** = boxes ticked / total boxes, exactly like the *Progress* column of the Excel. Example: Planned + RHS + LHS ticked, no FIR, no report = 3 of 5 = **60 %**.
- **Fleet % executed** of a countermeasure = trucks executed (Executed or Closed out) / applicable trucks. This is the *Fleet % complete* of the action plan.
- **Countermeasure status**: *Done* (100 % executed), *Overdue* (target date passed), *Due soon* (target within 45 days - configurable), *On track*, *No target* (no date set).

BAR COLOUR (GANTT AND PDF)	MEANING
Green	Done / executed.
Red	Overdue: the fleet target date has passed and the work is not finished.
Orange	In progress / pending, still within the target.

WORKING HOURS, DOWNTIME AND COMPLETION

Each countermeasure has an estimate of **hours per truck** (the *Duration per machine*, e.g. 12 h). The app uses it like this:

- Hours only count on trucks where the countermeasure is **not executed yet**. Executed, closed-out and N/A trucks count 0 h.
- If some execution steps are already done, only the open part counts. Example: Front Nose Cone = 12 h with steps RHS and LHS; RD186 has RHS done -> **6 h** remain.
- **Remaining work** of the project = sum of the remaining hours of every truck.
- **Estimated downtime of a truck** = sum of its remaining hours; shown also in days using *work hours per day* (default 12).
- **Time to 100 %** = remaining hours / (work hours per day x trucks worked in parallel). The **estimated completion** date is today + that number of days (calendar days).
- The estimated completion date is **based on continuous work on two trucks in parallel, without any delays** (default: 12 h per day, 2 trucks). Breakdowns, missing parts, holidays or trucks not released by operations will push the real date later. Admins can change the capacity (6.5).

EXAMPLE

Remaining work 3,999 h, 12 h per day, 2 trucks in parallel: 3,999 / (12 x 2) = 167 days -> estimated completion 29-Apr-2027, assuming continuous work without any delays. With only 1 truck at a time it would be 3,999 / 12 = 334 days (13-Oct-2027).

4. THE LANDING PAGE (EVERYONE)

The landing page is what everybody sees before signing in. From top to bottom:

4.1 FLEET KPI CARDS

CARD	MEANING
Fleet executed	% of all applicable truck-jobs that are executed. E.g. 55 % = 561 of 1027.
Fully closed out	% of truck-jobs with execution + FIR + report.
Countermeasures done	Countermeasures executed on every applicable truck, e.g. 1 / 20.
Past target date	Countermeasures whose fleet target has passed without finishing.

4.2 COUNTERMEASURE TIMELINE (FLEET GANTT)

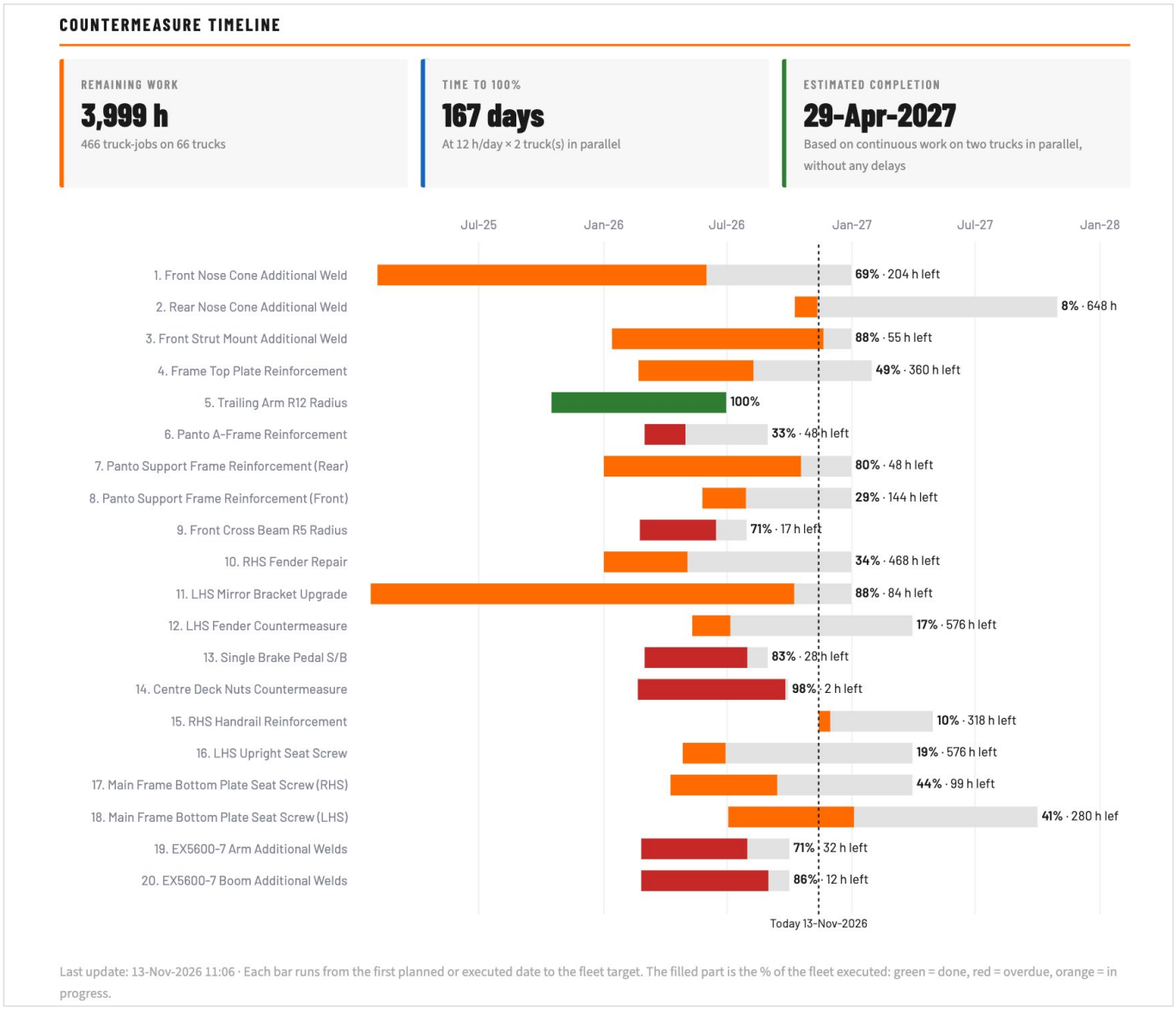


Figure 4. Countermeasure timeline with the forecast cards on top. Each bar = one countermeasure; the coloured part is the % of the fleet executed.

- Three cards on top: **Remaining work** (hours, truck-jobs, trucks), **Time to 100 %** (days at the configured capacity) and **Estimated completion** (date). The note under the date reminds that it is *based on continuous work on two trucks in parallel, without any delays*.
- Each grey bar runs from the first planned or executed date of the countermeasure to its **fleet target**. The coloured part shows the **% of the fleet executed** (green / red / orange as in chapter 3).
- At the end of the bar: the % and the hours still left, e.g. *71 % - 17 h left*.
- The dotted vertical line is **today**. Bars ending left of the line with orange or red still to go are late.
- Move the mouse over a bar (or tap it) to see start, target, trucks executed, status and hours.
- In light grey under the chart: **Last update** - the date and time of the last change saved by anybody.

EXAMPLE

Row *9. Front Cross Beam R5 Radius*: the bar is **red**, the label says **71 % - 17 h left**. Meaning: the fleet target (30-Jul-2026) has passed, 71 % of the trucks are executed and the 17 remaining trucks need about 17 h of work in total (1 h each).

4.3 TRUCK VIEW

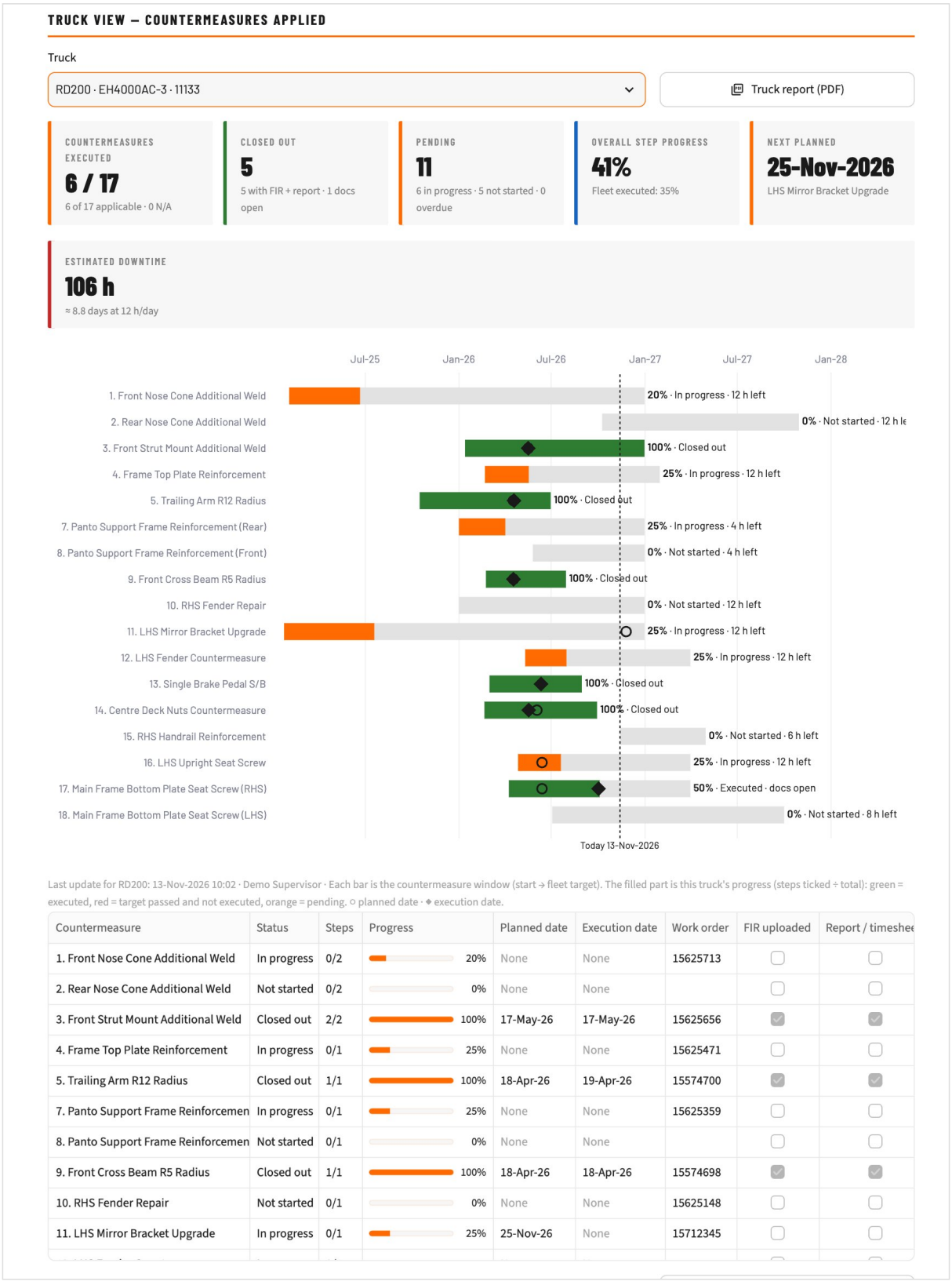


Figure 5. Truck view for RD200: scoreboard, estimated downtime, truck Gantt and detail table.

Pick a truck in the **Truck** box (you can type to search, e.g. *RD200*). The view shows:

CARD	EXAMPLE (RD200)	MEANING
Countermeasures executed	6 / 17	Executed out of applicable; N/A count below.
Closed out	5	Jobs with FIR and report; how many still have documents open.
Pending	11	In progress + not started; the card turns red if any is overdue.
Overall step progress	41 %	Average progress of its records (Excel method).
Next planned	25-Nov-2026	Next future planned date and its countermeasure.
Estimated downtime	106 h	Remaining hours of its pending work; ~8.8 days at 12 h/day.

- **Truck Gantt**: one bar per countermeasure that applies to this truck. The grey bar is the countermeasure window; the coloured part is **this truck's** progress. Green = executed, orange = pending, a **red label** = target passed and not executed.
- Markers: **circle** = planned date, **diamond** = execution date.
- Labels show progress, status and the hours left on that truck, e.g. *25 % - In progress - 12 h left*.
- In light grey: last update for this truck (when and who).
- **Detail table**: status, steps, progress, planned and execution dates, work order, FIR, report, target, hours per truck, remaining hours and note.
- **Truck report (PDF)** (button next to the truck selector): downloads a LANDCROS report of the selected truck - scoreboard, truck Gantt with planned / execution markers, and the detail table with notes. Useful for PM planning meetings and for the customer. Every copy gets a **Report ID** and is recorded in the Print register (6.7).

4.4 ACTION PLAN AND TOP PAGE

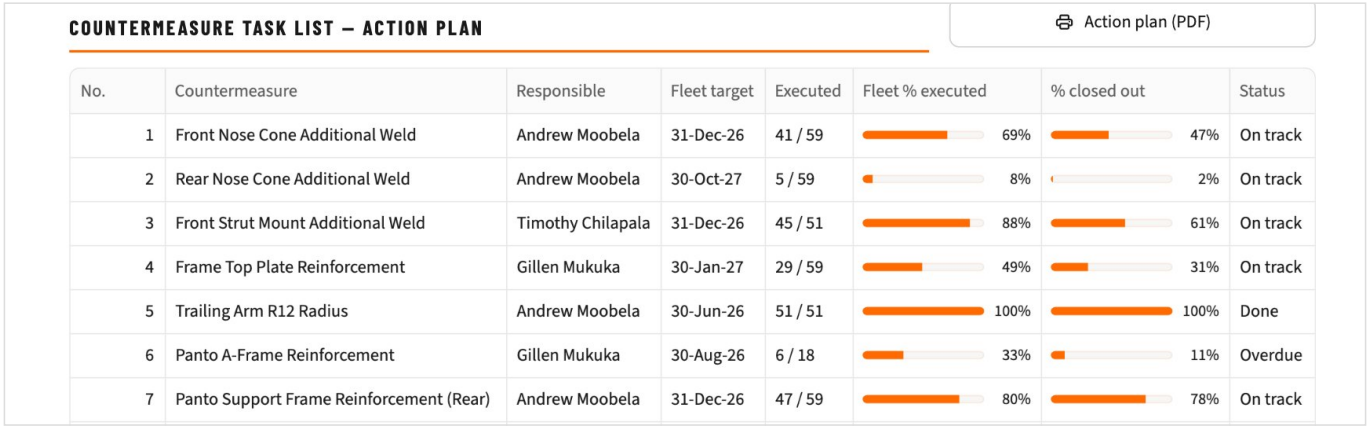


Figure 6. Countermeasure task list (action plan) with the Action plan (PDF) download button.

- **Countermeasure task list - action plan**: responsible, fleet target, executed / applicable, fleet % executed, % closed out and status. Click a column title to sort.
- **Action plan (PDF)**: downloads the LANDCROS action plan report. Page 1 is the **countermeasure Gantt** with remaining work, time to 100 % and estimated completion; then the KPIs, the countermeasure task list with fleet % bars, and the alerts. Each copy carries a **Report ID** in its footer and is recorded in the Print register (6.7).
- **Top page**: the truck x countermeasure matrix, coloured by status, like the *Top Page* sheet of the Excel. Scroll right to see all countermeasures.

TIP

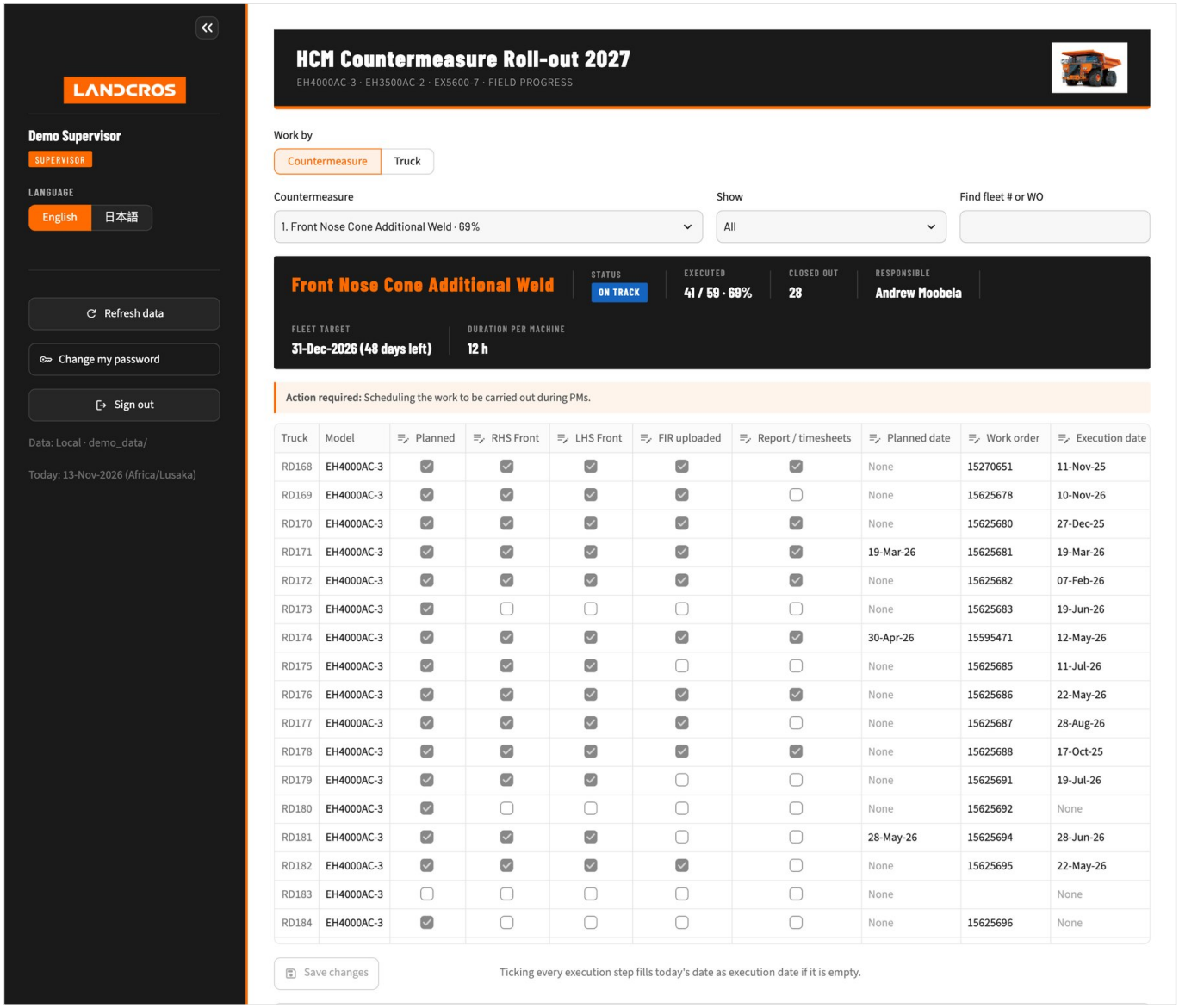
In the Top page the EX5600-7 excavators appear first and have empty cells for the truck countermeasures - that is normal: those countermeasures do not apply to them. Admins can filter the matrix by model (chapter 6).

5. RECORDING PROGRESS (SUPERVISORS)

After signing in, supervisors see the progress editor. You can work in two ways - choose with **Work by**:

WORK BY	USE IT WHEN
Countermeasure	You do the same job on many trucks (e.g. a welding campaign). One table = all the trucks of one countermeasure, with each execution step (RHS / LHS) as its own column.
Truck	A truck is in the workshop and you work on several countermeasures. One table = all the countermeasures of that truck, with a single Executed box per countermeasure.

5.1 WORKING BY COUNTERMEASURE



Truck	Model	Planned	RHS Front	LHS Front	FIR uploaded	Report / timesheets	Planned date	Work order	Execution date
RD168	EH4000AC-3	☑	☑	☑	☑	☑	None	15270651	11-Nov-25
RD169	EH4000AC-3	☑	☑	☑	☑	☐	None	15625678	10-Nov-26
RD170	EH4000AC-3	☑	☑	☑	☑	☑	None	15625680	27-Dec-25
RD171	EH4000AC-3	☑	☑	☑	☑	☑	19-Mar-26	15625681	19-Mar-26
RD172	EH4000AC-3	☑	☑	☑	☑	☑	None	15625682	07-Feb-26
RD173	EH4000AC-3	☑	☐	☐	☐	☐	None	15625683	19-Jun-26
RD174	EH4000AC-3	☑	☑	☑	☑	☑	30-Apr-26	15595471	12-May-26
RD175	EH4000AC-3	☑	☑	☑	☐	☐	None	15625685	11-Jul-26
RD176	EH4000AC-3	☑	☑	☑	☑	☑	None	15625686	22-May-26
RD177	EH4000AC-3	☑	☑	☑	☑	☐	None	15625687	28-Aug-26
RD178	EH4000AC-3	☑	☑	☑	☑	☑	None	15625688	17-Oct-25
RD179	EH4000AC-3	☑	☑	☑	☐	☐	None	15625691	19-Jul-26
RD180	EH4000AC-3	☑	☐	☐	☐	☐	None	15625692	None
RD181	EH4000AC-3	☑	☑	☑	☐	☐	28-May-26	15625694	28-Jun-26
RD182	EH4000AC-3	☑	☑	☑	☑	☐	None	15625695	22-May-26
RD183	EH4000AC-3	☐	☐	☐	☐	☐	None		None
RD184	EH4000AC-3	☑	☐	☐	☐	☐	None	15625696	None

Figure 7. Supervisor view, working by countermeasure: selector, filters, countermeasure badge and the editable table.

1. Choose the **Countermeasure** (the list shows its fleet %).
2. Optional: **Show** filters by status (e.g. only *Not started*); **Find fleet # or WO** searches by truck number or work order.

3. The black badge shows the status, trucks executed, closed out, responsible, fleet target (days left), duration per machine and the action required.
4. Click the boxes and cells in the table to change them. Changed rows are counted under the table: *"2 rows changed - not saved yet"*.
5. Press **Save changes**. A green message confirms *Saved N rows* and the table, KPIs and charts update.

COLUMN	HOW TO FILL IT
Planned	Tick when the job is scheduled.
Step columns (e.g. RHS Front, LHS Front)	Tick each part when it is physically done.
FIR uploaded	Tick when the Field Investigation Report is uploaded.
Report / timesheets	Tick when the report and timesheets are filed.
Planned date	Double-click, pick the date. Ticks Planned automatically.
Work order	Type the KMP work order number.
Execution date	Filled with today automatically when all steps are ticked (if empty). You can change it to the real date.
Note	Free text, e.g. Waiting for parts.
N/A	Greyed out for supervisors - only admins can change it.

EXAMPLE

Example 1 - one side welded. On 12-Nov the night shift welded the RHS of the front nose cone on RD173 (WO 15625683).
1) Work by Countermeasure *-> 1. Front Nose Cone Additional Weld*. 2) Type *RD173* in Find fleet. 3) Tick **RHS Front**, type the WO, note *LHS next PM*. 4) Save changes.
Result: status *In progress*, remaining hours for RD173 drop from 12 h to 6 h.

EXAMPLE

Example 2 - finishing the job. Two days later LHS is welded. Tick **LHS Front** and Save. Because all steps are now ticked, *Execution date* is filled with today and the status becomes *Executed - docs open*. An alert *Executed, documents missing* appears for the admins. When the FIR and the report are filed, tick **FIR uploaded** and **Report / timesheets** and Save *-> Closed out*.

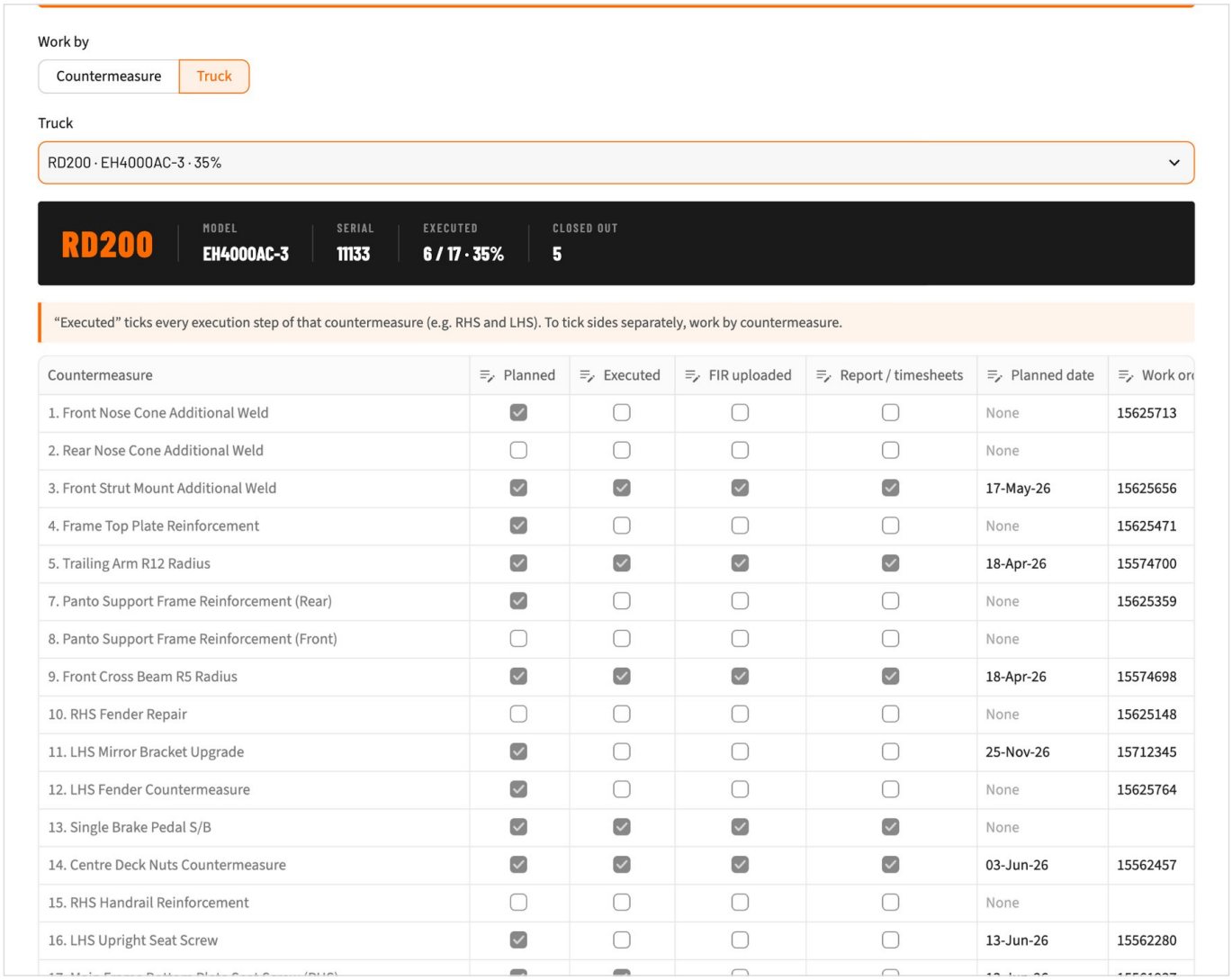
EXAMPLE

Example 3 - scheduling. RD188 is booked for the rear nose cone on 25-Nov with WO 15712345. Choose *2. Rear Nose Cone*, find RD188, set **Planned date** = 25-Nov-2026 and the work order, Save. *Planned* is ticked automatically and the date appears as a circle in the truck Gantt and as *Next planned* in the truck scoreboard. If 25-Nov passes without execution, an alert *Planned date passed, not executed* is raised.

IMPORTANT

Changing the countermeasure, the **Show** filter or the **Find** text **discards the changes you have not saved**. Always press **Save changes** before switching.

5.2 WORKING BY TRUCK



Countermeasure	Planned	Executed	FIR uploaded	Report / timesheets	Planned date	Work or
1. Front Nose Cone Additional Weld	☑	☐	☐	☐	None	15625713
2. Rear Nose Cone Additional Weld	☐	☐	☐	☐	None	
3. Front Strut Mount Additional Weld	☑	☑	☑	☑	17-May-26	15625656
4. Frame Top Plate Reinforcement	☑	☐	☐	☐	None	15625471
5. Trailing Arm R12 Radius	☑	☑	☑	☑	18-Apr-26	15574700
7. Panto Support Frame Reinforcement (Rear)	☑	☐	☐	☐	None	15625359
8. Panto Support Frame Reinforcement (Front)	☐	☐	☐	☐	None	
9. Front Cross Beam R5 Radius	☑	☑	☑	☑	18-Apr-26	15574698
10. RHS Fender Repair	☐	☐	☐	☐	None	15625148
11. LHS Mirror Bracket Upgrade	☑	☐	☐	☐	25-Nov-26	15712345
12. LHS Fender Countermeasure	☑	☐	☐	☐	None	15625764
13. Single Brake Pedal S/B	☑	☑	☑	☑	None	
14. Centre Deck Nuts Countermeasure	☑	☑	☑	☑	03-Jun-26	15562457
15. RHS Handrail Reinforcement	☐	☐	☐	☐	None	
16. LHS Upright Seat Screw	☑	☐	☐	☐	13-Jun-26	15562280

Figure 8. Supervisor view, working by truck: truck badge and one row per countermeasure.

1. Choose **Work by -> Truck** and pick the truck (e.g. *RD200 - EH4000AC-3 - 35 %*).
2. Each row is one countermeasure. **Executed** ticks **every** execution step of that countermeasure at once (e.g. both RHS and LHS).
3. Fill planned date, work order, FIR, report and notes as above, then **Save changes**.

TIP

To tick only one side (RHS or LHS) of a two-step countermeasure, use *Work by -> Countermeasure*: the truck view only has one Executed box.

EXAMPLE

Example 4 - truck in for PM. RD200 is in the workshop for its PM and the team does the rear and front panto support frame reinforcements. Work by Truck -> RD200, tick **Executed** on rows 7 and 8, Save. RD200's estimated downtime drops by 8 h (4 h each).

5.3 CHANGE HISTORY

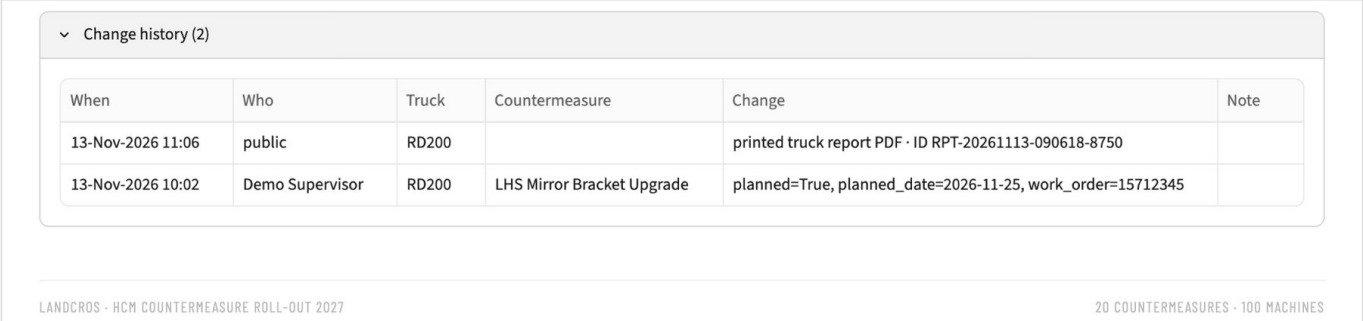


Figure 9. Change history under the table: when, who, truck, countermeasure, what changed and the note.

Open **Change history** under the table to see every saved change for that countermeasure or truck: date and time, who, and exactly what changed (e.g. *planned=True, planned_date=2026-11-25, work_order=15712345*). Nothing is ever deleted - if a value was wrong, simply save the correct value; the latest one wins.

6. MANAGING THE PROJECT (ADMINS)

Admins see eight tabs. The master user sees the same tabs when *View* is set to *Admin* in the sidebar.

TAB	USE IT TO
Summary	See the KPIs, all open alerts, the action plan and the latest changes.
Top page	See the truck x countermeasure matrix, filtered by model.
Progress	Record progress like a supervisor, and set N/A.
Countermeasures	Edit the action plan; add or remove countermeasures.
Working hours	Set the hours per truck of each countermeasure and the workshop capacity; see downtime per truck.
Trucks	See the fleet; add or remove trucks.
Reports	Download Excel / PDF / CSV and publish them.
Settings	Time zone, due-soon warning, your password.

6.1 SUMMARY AND ALERTS

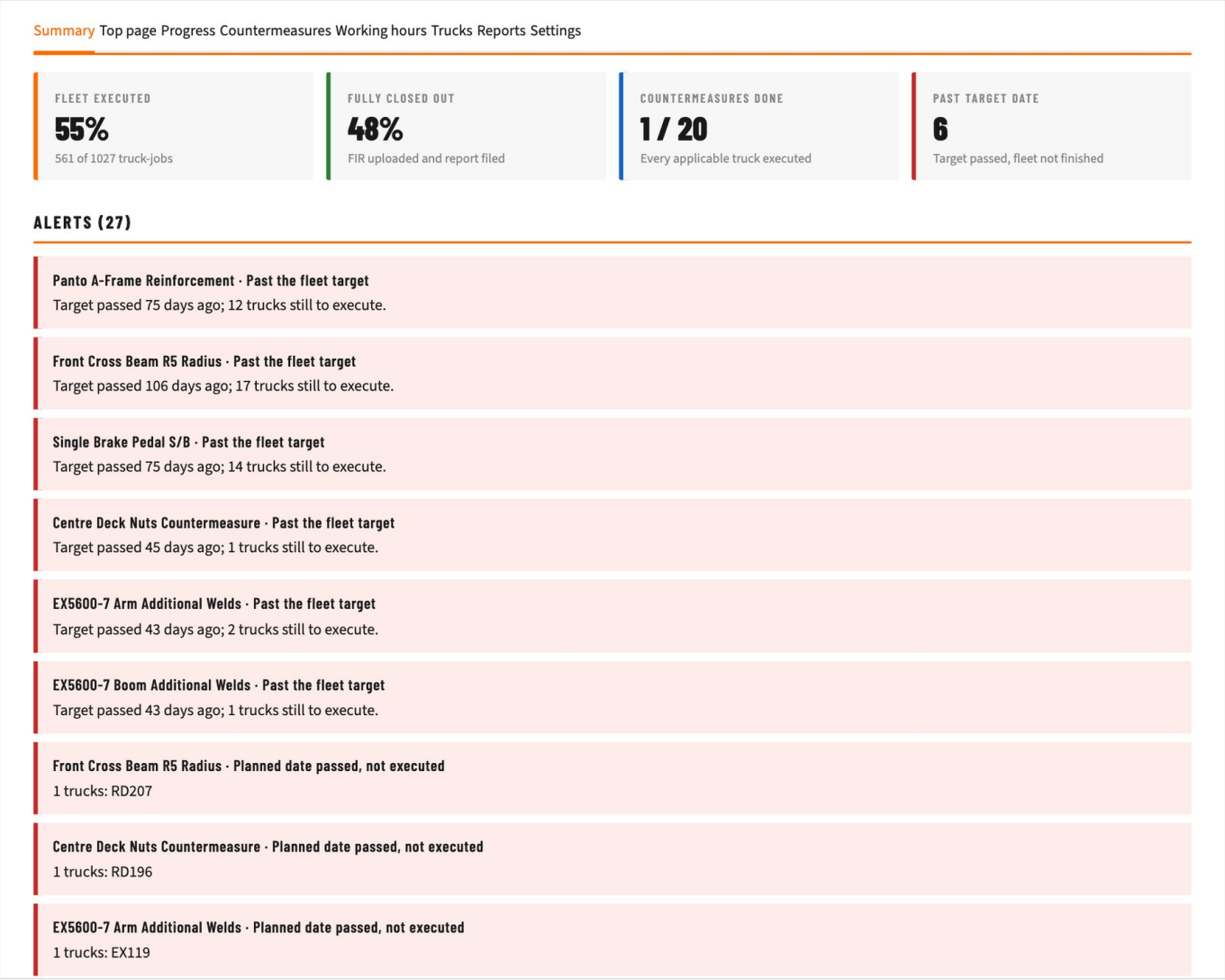
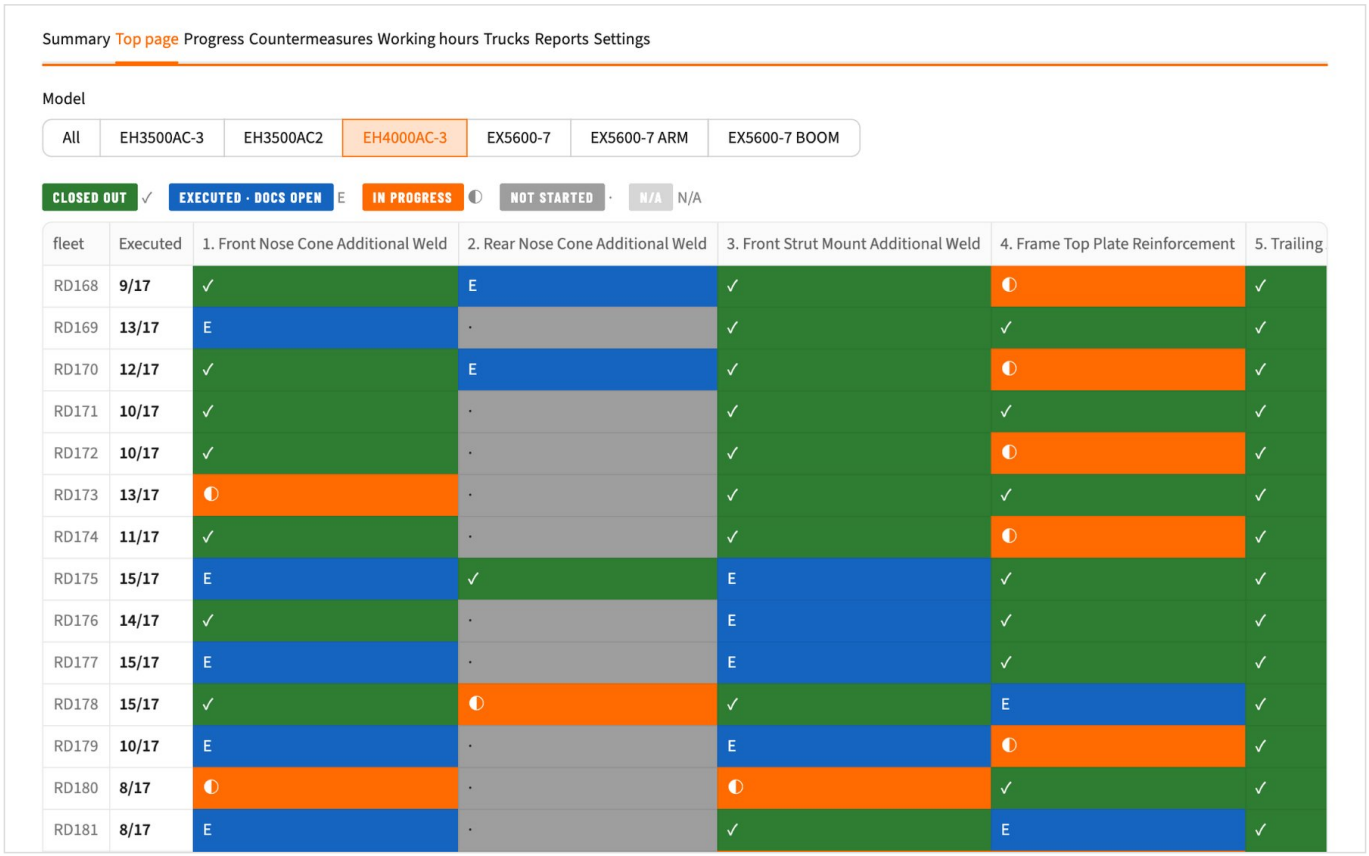


Figure 10. Summary tab: KPI cards and the alert list (most urgent first).

ALERT	RAISED WHEN	TYPICAL ACTION
Past the fleet target	Target date passed and trucks still to execute.	Re-plan the work or agree a new target (Countermeasures tab).
Planned date passed, not executed	A truck's planned date is in the past and it is not executed.	Ask the supervisor to execute it or set a new planned date.
Fleet target approaching	Target within the due-soon window (45 days by default).	Check that the remaining trucks are scheduled.
Executed, documents missing	Executed trucks without FIR or report.	Chase the FIR / report and tick them.

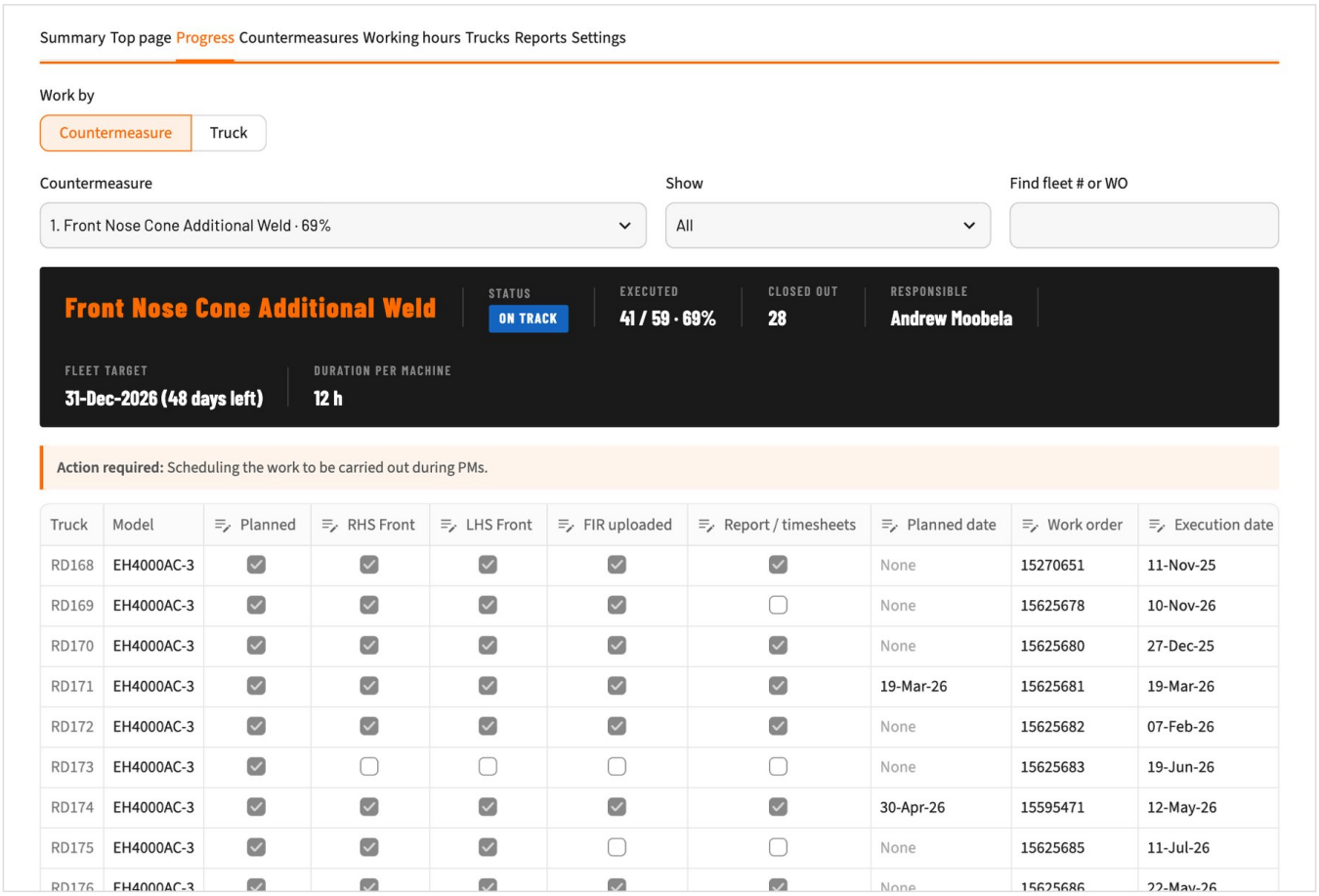
6.2 TOP PAGE BY MODEL



fleet	Executed	1. Front Nose Cone Additional Weld	2. Rear Nose Cone Additional Weld	3. Front Strut Mount Additional Weld	4. Frame Top Plate Reinforcement	5. Trailing
RD168	9/17	✓	E	✓	◐	✓
RD169	13/17	E	·	✓	✓	✓
RD170	12/17	✓	E	✓	◐	✓
RD171	10/17	✓	·	✓	✓	✓
RD172	10/17	✓	·	✓	◐	✓
RD173	13/17	◐	·	✓	✓	✓
RD174	11/17	✓	·	✓	◐	✓
RD175	15/17	E	✓	E	✓	✓
RD176	14/17	✓	·	E	✓	✓
RD177	15/17	E	·	E	✓	✓
RD178	15/17	✓	◐	✓	E	✓
RD179	10/17	E	·	E	◐	✓
RD180	8/17	◐	·	◐	✓	✓
RD181	8/17	E	·	✓	E	✓

Figure 11. Top page filtered to EH4000AC-3. Use the Model buttons to switch.

6.3 PROGRESS AND N/A



Truck	Model	Planned	RHS Front	LHS Front	FIR uploaded	Report / timesheets	Planned date	Work order	Execution date
RD168	EH4000AC-3	☑	☑	☑	☑	☑	None	15270651	11-Nov-25
RD169	EH4000AC-3	☑	☑	☑	☑	☐	None	15625678	10-Nov-26
RD170	EH4000AC-3	☑	☑	☑	☑	☑	None	15625680	27-Dec-25
RD171	EH4000AC-3	☑	☑	☑	☑	☑	19-Mar-26	15625681	19-Mar-26
RD172	EH4000AC-3	☑	☑	☑	☑	☑	None	15625682	07-Feb-26
RD173	EH4000AC-3	☑	☐	☐	☐	☐	None	15625683	19-Jun-26
RD174	EH4000AC-3	☑	☑	☑	☑	☑	30-Apr-26	15595471	12-May-26
RD175	EH4000AC-3	☑	☑	☑	☐	☐	None	15625685	11-Jul-26
RD176	EH4000AC-3	☑	☑	☑	☑	☑	None	15625686	22-May-26

Figure 12. Progress tab - the same editor as supervisors, with the N/A column enabled.

EXAMPLE

Example 5 - N/A. In the Excel, RD220 had no status for the front strut mount, so it was loaded as **N/A** (not applicable). Now the truck does need the weld. Progress -> Countermeasure *3. Front Strut Mount Additional Weld*, Show *N/A*, find RD220, **untick N/A**, Save. RD220 now counts in the strut's %, hours and alerts as *Not started*. To exclude a truck, do the opposite: tick N/A and Save.

6.4 COUNTERMEASURES: EDIT, ADD, REMOVE

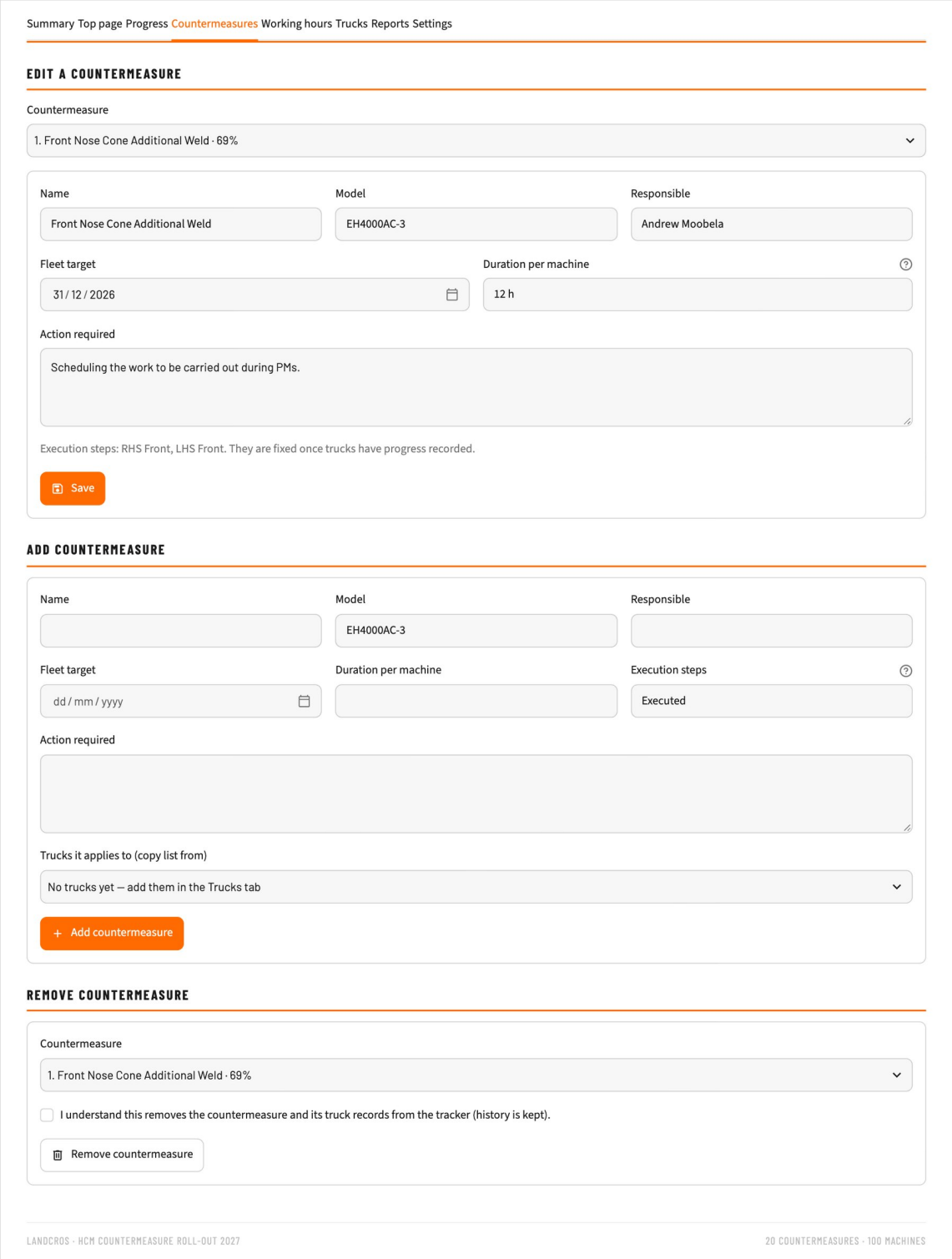


Figure 13. Countermeasures tab: Edit a countermeasure, Add countermeasure and Remove countermeasure.

Edit a countermeasure

Pick the countermeasure, change name, model, responsible, fleet target, duration per machine or action required, and press **Save**. Only the fields you changed are recorded.

EXAMPLE

Example 6 - new target date. The customer agreed to move *12. LHS Fender Countermeasure* to 30-Apr-2027. Select it, change **Fleet target** to 30/04/2027, Save. Its status, the alerts, the Gantt bar and the PDF update at once.

Add a countermeasure

1. Fill **Name** (required), model, responsible, fleet target, duration (e.g. *10 h*) and action required.
2. **Execution steps**: leave *Executed* for a single step, or type the steps separated by commas, e.g. *RHS, LHS*.
3. **Trucks it applies to (copy list from)**: copy the truck list (including N/A trucks) of a similar countermeasure, or choose *No trucks yet* and add them later in the Trucks tab.
4. Press **Add countermeasure**.

EXAMPLE

Example 7 - new weld. Name *Rear Axle Housing Additional Weld*, model *EH4000AC-3*, responsible *Andrew Moobela*, target 30/06/2027, duration *10 h*, steps *RHS, LHS*, copy trucks from *1. Front Nose Cone*. It appears at the end of the action plan with 0 %, and adds 10 h x the number of applicable trucks to the remaining work.

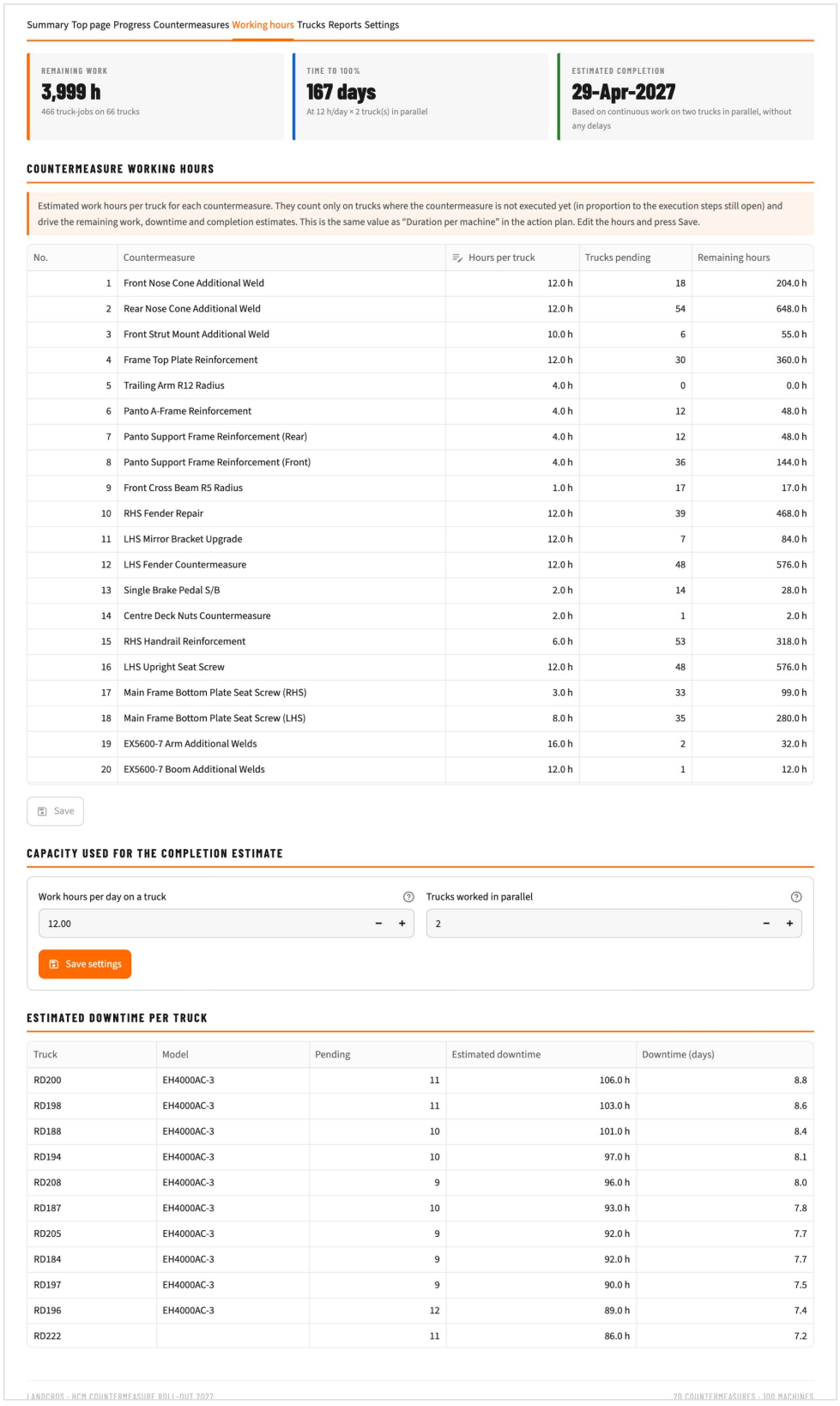
IMPORTANT

The **execution steps cannot be changed** after the countermeasure is created. Check them before pressing Add.

Remove a countermeasure

Choose it, tick the confirmation box and press **Remove countermeasure**. It disappears from the tracker and reports; its change history is kept.

6.5 WORKING HOURS



Summary Top page Progress Countermeasures Working hours Trucks Reports Settings

REMAINING WORK
3,999 h
466 truck-jobs on 66 trucks

TIME TO 100%
167 days
At 12 h/day × 2 truck(s) in parallel

ESTIMATED COMPLETION
29-Apr-2027
Based on continuous work on two trucks in parallel, without any delays

COUNTERMEASURE WORKING HOURS

Estimated work hours per truck for each countermeasure. They count only on trucks where the countermeasure is not executed yet (in proportion to the execution steps still open) and drive the remaining work, downtime and completion estimates. This is the same value as "Duration per machine" in the action plan. Edit the hours and press Save.

No.	Countermeasure	Hours per truck	Trucks pending	Remaining hours
1	Front Nose Cone Additional Weld	12.0 h	18	204.0 h
2	Rear Nose Cone Additional Weld	12.0 h	54	648.0 h
3	Front Strut Mount Additional Weld	10.0 h	6	55.0 h
4	Frame Top Plate Reinforcement	12.0 h	30	360.0 h
5	Trailing Arm R12 Radius	4.0 h	0	0.0 h
6	Panto A-Frame Reinforcement	4.0 h	12	48.0 h
7	Panto Support Frame Reinforcement (Rear)	4.0 h	12	48.0 h
8	Panto Support Frame Reinforcement (Front)	4.0 h	36	144.0 h
9	Front Cross Beam R5 Radius	1.0 h	17	17.0 h
10	RHS Fender Repair	12.0 h	39	468.0 h
11	LHS Mirror Bracket Upgrade	12.0 h	7	84.0 h
12	LHS Fender Countermeasure	12.0 h	48	576.0 h
13	Single Brake Pedal S/B	2.0 h	14	28.0 h
14	Centre Deck Nuts Countermeasure	2.0 h	1	2.0 h
15	RHS Handrail Reinforcement	6.0 h	53	318.0 h
16	LHS Upright Seat Screw	12.0 h	48	576.0 h
17	Main Frame Bottom Plate Seat Screw (RHS)	3.0 h	33	99.0 h
18	Main Frame Bottom Plate Seat Screw (LHS)	8.0 h	35	280.0 h
19	EX5600-7 Arm Additional Welds	16.0 h	2	32.0 h
20	EX5600-7 Boom Additional Welds	12.0 h	1	12.0 h

Save

CAPACITY USED FOR THE COMPLETION ESTIMATE

Work hours per day on a truck: 12.00
Trucks worked in parallel: 2
Save settings

ESTIMATED DOWNTIME PER TRUCK

Truck	Model	Pending	Estimated downtime	Downtime (days)
RD200	EH4000AC-3	11	106.0 h	8.8
RD198	EH4000AC-3	11	103.0 h	8.6
RD188	EH4000AC-3	10	101.0 h	8.4
RD194	EH4000AC-3	10	97.0 h	8.1
RD208	EH4000AC-3	9	96.0 h	8.0
RD187	EH4000AC-3	10	93.0 h	7.8
RD205	EH4000AC-3	9	92.0 h	7.7
RD184	EH4000AC-3	9	92.0 h	7.7
RD197	EH4000AC-3	9	90.0 h	7.5
RD196	EH4000AC-3	12	89.0 h	7.4
RD222		11	86.0 h	7.2

LANDCROS - HCM COUNTERMEASURE ROLL-OUT 2027 — 20 COUNTERMEASURES · 100 MACHINES

Figure 14. Working hours tab: forecast cards, hours per truck table, capacity and estimated downtime per truck.

- **Countermeasure working hours** table: type the **Hours per truck** of each countermeasure (e.g. 12.0) and press **Save**. *Trucks pending* and *Remaining hours* are calculated for you.
- These hours are the same value as *Duration per machine*: changing one changes the other.
- **Capacity used for the completion estimate**: *Work hours per day on a truck* (e.g. 12 for one 12-hour shift; default 12) and *Trucks worked in parallel* (default 2). Press **Save settings**. The note under *Estimated completion* always shows the number of trucks used.
- **Estimated downtime per truck**: trucks with the most outstanding work first, in hours and days.
- A yellow warning lists countermeasures without hours - their trucks are not counted until you set them.

EXAMPLE

Example 8 - longer job than planned. The mirror bracket upgrade really takes 20 h, not 12 h. Change *11. LHS Mirror Bracket Upgrade* from 12.0 to 20.0 and Save. With 7 trucks pending, its remaining hours go from 84 h to 140 h, and the project's remaining work, time to 100 % and completion date move accordingly.

EXAMPLE

Example 9 - third bay. From December three trucks can be worked at the same time. Set *Trucks worked in parallel* = 3 and Save settings: *Time to 100 %* drops from 167 to 112 days, the estimated completion moves from 29-Apr-2027 to 05-Mar-2027, and the note reads *based on continuous work on three trucks in parallel, without any delays*.

6.6 TRUCKS

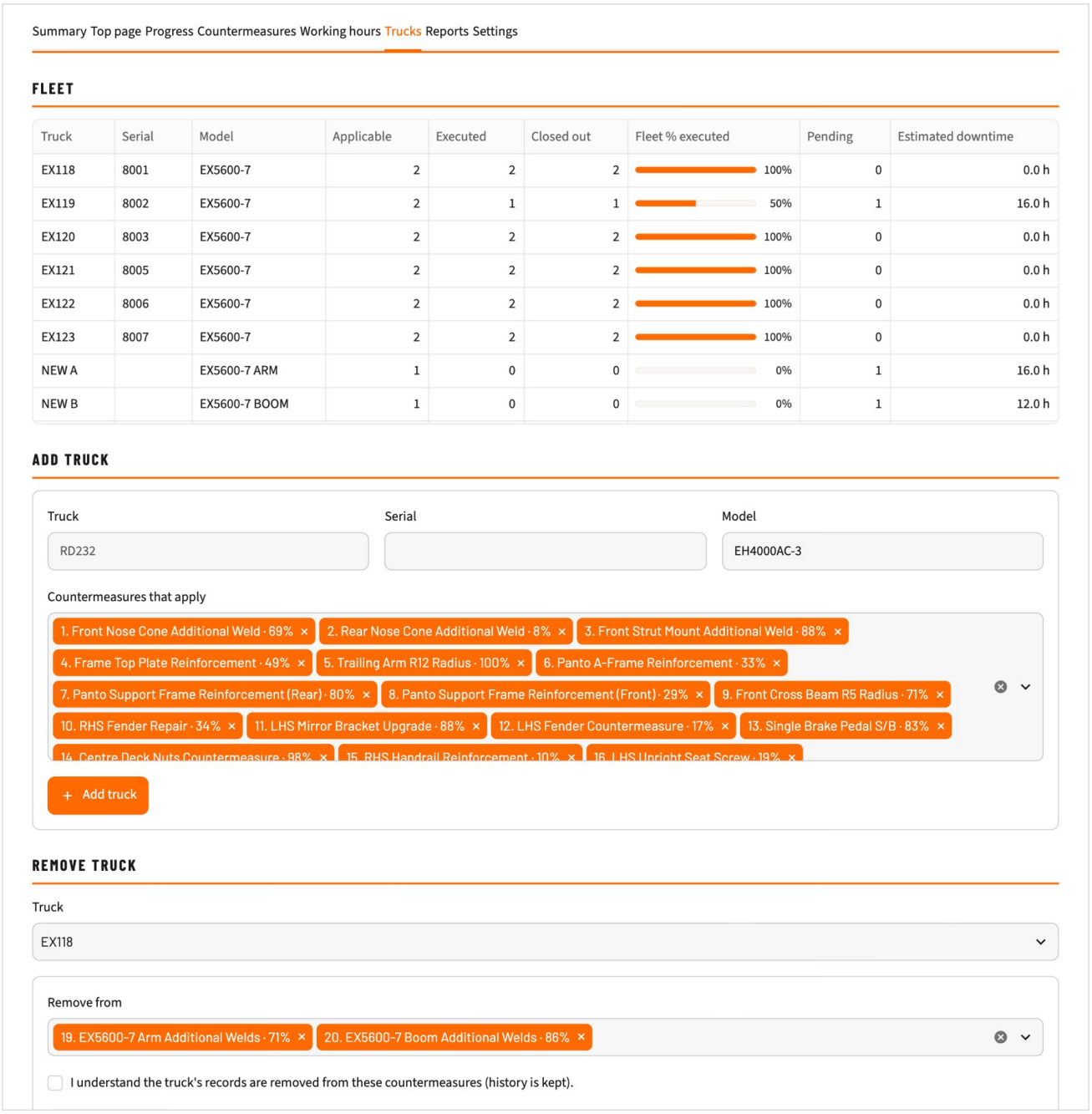


Figure 15. Trucks tab: fleet table (with pending countermeasures and estimated downtime), Add truck and Remove truck.

EXAMPLE

Example 10 - new truck. RD232 (serial 50189, EH4000AC-3) joins the fleet. Trucks -> Add truck: Truck *RD232*, Serial *50189*, Model *EH4000AC-3*, keep the countermeasures that apply (remove the ones that do not), press **Add truck**. It appears in every table as *Not started*.

To remove a truck from some or all countermeasures, choose it in **Remove truck**, keep the countermeasures to remove it from, tick the confirmation box and press **Remove truck**. To exclude a truck from one countermeasure without deleting it, use **N/A** instead (6.3).

6.7 REPORTS AND SETTINGS

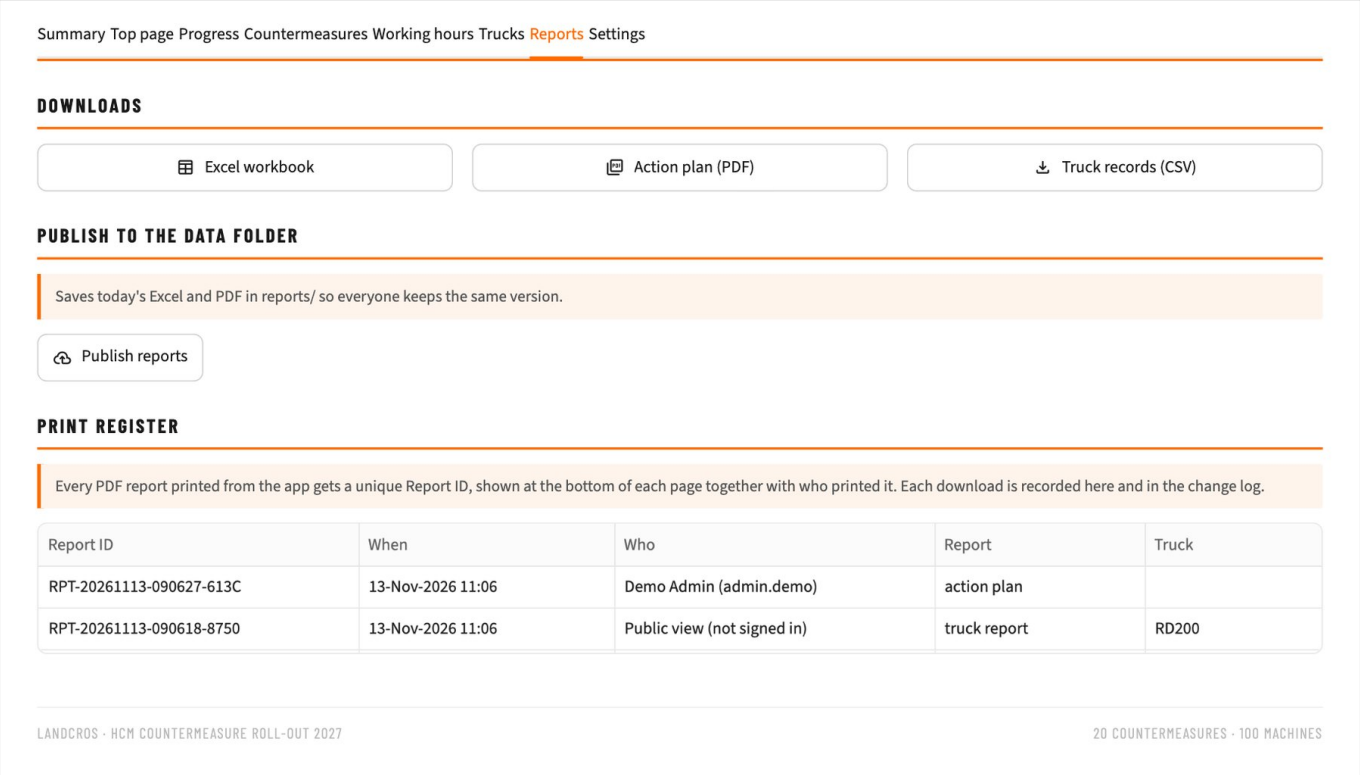


Figure 16. Reports tab: downloads, Publish reports and the Print register.

BUTTON	YOU GET
Excel workbook	Summary, Countermeasures (incl. hours), Top Page, Truck records, Trucks (incl. downtime) and Change log sheets.
Action plan (PDF)	The LANDCROS action plan report: page 1 countermeasure Gantt with the hours forecast, then KPIs, task list with fleet % bars and alerts.
Truck records (CSV)	Every truck record in one file, for other tools.
Publish reports	Saves today's Excel and PDF in the reports/ folder of the data repository, so everybody keeps the same version.

Print register and Report ID

Every PDF report printed from the app - the **Action plan (PDF)** (landing page or Reports tab) and the **Truck report (PDF)** (truck view) - gets a unique **Report ID**, for example *RPT-20261113-090627-613C* (date, time and a random code). The ID and the name of who printed it are shown at the bottom of every page of the PDF.

Figure 17. Footer of a printed action plan: Report ID and Printed by.

- Each download is recorded with its Report ID, report type, truck (for truck reports), date and time, and the user. Downloads from the public landing page are recorded as *Public view (not signed in)*.
- Admins see the list in **Reports -> Print register**; it also appears in *Recent changes*, in the truck's *Change history* and in the *Change log* sheet of the Excel workbook.
- To check a paper copy, look up its Report ID in the Print register: you will see who printed it and when.

EXAMPLE

Example 11 - which copy is this? At the weekly meeting a printed truck report for RD200 shows *Report ID RPT-20261113-090618-8750*. In Reports -> Print register that ID reads *13-Nov-2026 11:06 - Public view (not signed in) - truck report - RD200*, so everyone knows it shows the status at 11:06 on 13 November.

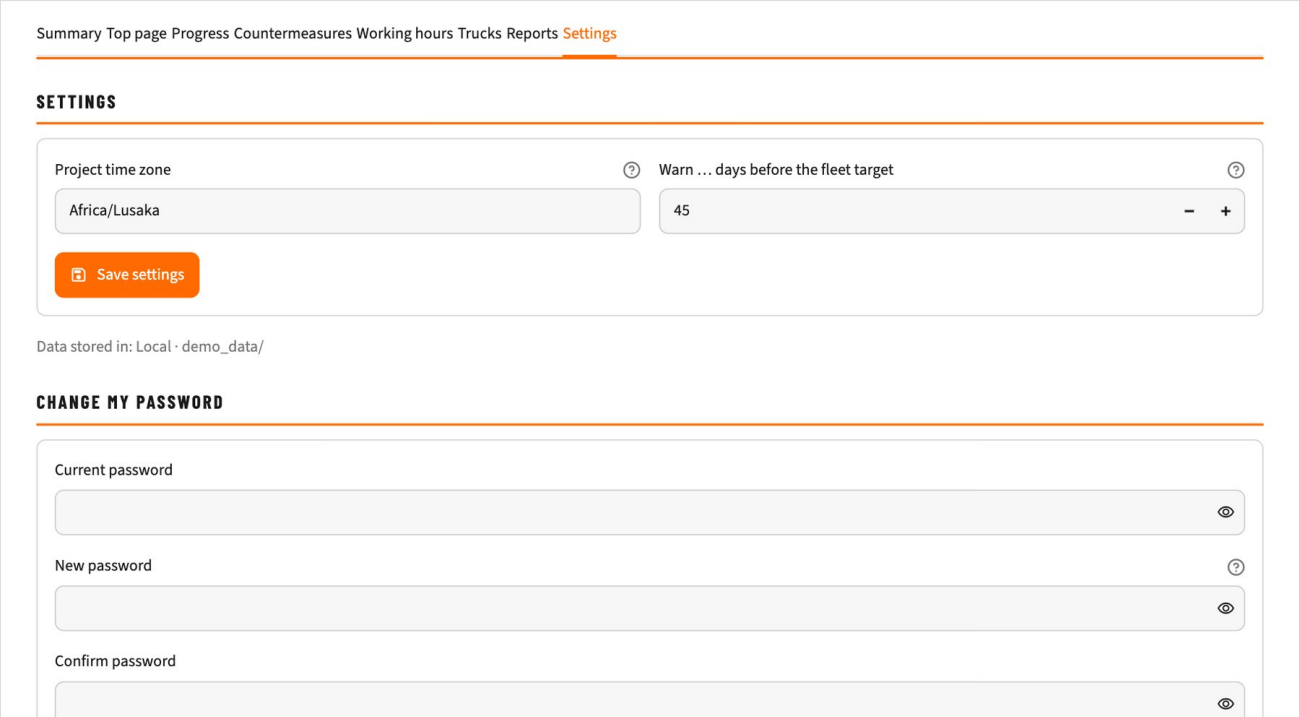


Figure 18. Settings tab: time zone, due-soon warning and Change my password.

- **Project time zone** defines *today* for every date calculation (default *Africa/Lusaka*).
- **Warn ... days before the fleet target** sets the *Due soon* window (default 45 days).
- **Change my password** (admins): current password, new password (8+ characters) twice, Save.

7. USER ACCOUNTS (APP OWNER)

Accounts are created only by the app owner, from a terminal on the owner's computer, in the project folder. Passwords are typed without being shown and are stored encrypted (bcrypt).

COMMAND	WHAT IT DOES
python scripts/users.py add	Create a user: username, full name, role (admin / supervisor), password.
python scripts/users.py list	List users, roles and whether they are active.
python scripts/users.py password <user>	Reset a forgotten password.
python scripts/users.py role <user> admin	Change a role (admin or supervisor).
python scripts/users.py rename <user> "Name"	Change the full name shown in the app.
python scripts/users.py deactivate <user>	Remove access but keep the user's history.
python scripts/users.py activate <user>	Give access back.
python scripts/users.py master-hash	Create the [master] block for the secrets (owner login).

IMPORTANT

After running **master-hash**, paste the new [master] block into *.streamlit/secrets.toml* **and** into Streamlit Cloud -> App settings -> Secrets, then **Reboot** the online app (chapter 9). Otherwise the new password does not work online.

TIP

A user who leaves the project should be **deactivated**, not deleted, so their name stays next to the changes they made. Deactivation takes effect within about 30 seconds.

ACCOUNTS IN USE

USERNAME	NAME	ROLE	NOTES
iibanez	Ivan Ibanez	Master (app owner)	Defined in the app secrets. Admin and Supervisor views.
marcos1	Marcos	Admin	Created 13-Nov-2026. Can sign in as Marcos1 or marcos1.
peter1	Peter	Admin	Created 13-Nov-2026. Can sign in as Peter 1, Peter1 or peter1.

IMPORTANT

Passwords are never written in this manual. They are given to each person privately by the app owner. Each admin should change the initial password the first day: Settings -> Change my password.

- **Usernames** ignore capital letters and spaces: *Peter 1*, *PETER1* and *peter1* are the same account.
- **Passwords** are case-sensitive: *Example123* is not the same as *example123*.

8. WHERE THE DATA IS SAVED

- The Excel tracker was only used **once**, to load the initial status. The app never writes to it.
- Every save creates a small, permanent file in the **data** branch of the GitHub repository (*iibanez828-droid/HCMZ-Countermeasures-project-EH4000AC3*). Nothing is overwritten, so the full history of who changed what and when is kept.
- The program code lives in the **main** branch. Saving data never restarts the app.
- The online app and any local copy that uses the same secrets work on the **same data**.
- Need an up-to-date Excel? Admin -> Reports -> **Excel workbook**.

> **IMPORTANT**
> Never edit or push files to the **data** branch by hand. Use the app.

9. REFRESH, REBOOT AND TROUBLESHOOTING

9.1 THE THREE LEVELS OF "REFRESH"

LEVEL	WHEN TO USE IT	HOW
1. Refresh data (sidebar)	You do not see a change another person saved; numbers look old.	Click Refresh data in the sidebar. The app also refreshes by itself every 30 seconds.
2. Reload the browser page	The page looks frozen, half-drawn or shows a red error box.	Press F5 (Windows) or Cmd+R (Mac). You will be signed out - sign in again. Unsaved table edits are lost.
3. Reboot the app (owner / admin with Streamlit access)	A new version was pushed but does not show; secrets were changed; the app is stuck or errors for everybody.	share.streamlit.io -> your app -> (three dots) -> Reboot app. Or, on the app page, Manage app (bottom right) -> (three dots) -> Reboot app. Wait 1-2 minutes.

> **TIP**
> Try the levels in order: Refresh data first, then reload the page, and reboot only if the first two do not help. Rebooting is safe - no data is lost, because the data is in GitHub - but everybody connected is disconnected for a minute.

9.2 AFTER THE OWNER CHANGES THE APP

- **Code changes** pushed to the *main* branch are picked up automatically by Streamlit Cloud within about 1-2 minutes. If after 5 minutes the change is still not visible: reload the page; if still not, **Reboot app**.
- **Secrets changes** (token, master password) are only read when the app starts: always **Reboot app** after saving the secrets.
- **Local copy** (owner's computer): stop it with **Ctrl+C** in the terminal and start it again with *streamlit run app.py*. Tip: start it with *streamlit run app.py --server.runOnSave true* so the page reloads by itself every time a file is saved.

9.3 COMMON PROBLEMS

WHAT YOU SEE	WHY	WHAT TO DO
"This app has gone to sleep due to inactivity"	Free Streamlit apps sleep after some hours without visitors.	Click Yes, get this app back up! and wait 30-60 s.
"The project has not been loaded yet" and sign-in fails	The online app has no secrets, so it cannot reach the data.	Owner: paste the secrets in Streamlit Cloud -> Settings -> Secrets, Save, Reboot app. Sidebar Data must say GitHub ...@data.
"Incorrect username or password"	Typo, caps, wrong account, account deactivated, or the master block in the secrets is outdated.	Check spelling (usernames are lower case). Ask the owner to reset the password or reboot after updating the secrets.
Somebody else's change is not visible	Your screen keeps data for up to 30 seconds.	Click Refresh data.
My edits disappeared	The countermeasure, filter or search was changed before saving, or the page was reloaded.	Enter them again and press Save changes first.
"Could not save. GitHub ... 401"	The GitHub token expired or is wrong.	Owner: create a new token, update the secrets, Reboot app.

"Could not save. GitHub ... 403 / 404"	Token without write permission, repository renamed, or GitHub limit reached.	Wait a minute and retry; if it persists, owner checks the token and repository.
Saving many rows is slow	Each row is saved separately to GitHub.	Wait for the confirmation; save in smaller batches (20-30 rows).
A pushed change does not appear online	Streamlit has not redeployed yet.	Wait 2 minutes, reload the page, then Reboot app.
Red error box after an update	Usually a temporary problem during a redeploy.	Reload the page. If it stays, owner opens Manage app to read the log and reboots.

> **NOTE**
> When you report a problem to the app owner, include: what you clicked, the exact message (a screenshot is best), the truck and countermeasure, and the time.

10. QUICK REFERENCE

SUPERVISOR - DAILY ROUTINE

1. Open the app link -> Sign in.
2. Work by **Countermeasure** (campaigns) or **Truck** (truck in the workshop).
3. Tick what was done today; enter planned dates, work orders and notes.
4. Press **Save changes** and wait for *Saved N rows*.
5. When FIR and report are filed, tick them and Save -> *Closed out*.
6. Click **Refresh data** if you expect to see a colleague's change. **Sign out** when done.

ADMIN - WEEKLY ROUTINE

1. Summary: review alerts (overdue, planned date passed, due soon, documents missing).
2. Countermeasures: update responsible, target dates and action required after the planning meeting.
3. Working hours: check hours per truck and capacity; read the estimated completion date.
4. Trucks / Progress: add new trucks; set N/A where a countermeasure does not apply.
5. Reports: download or **Publish reports** for the weekly meeting.

REMEMBER

- Nothing is deleted: correct a mistake by saving the right value.
- Ticking all execution steps fills the execution date with today (if empty) and ticks Planned.
- Only admins can set N/A. Execution steps are fixed once a countermeasure is created.
- Refresh data -> reload page -> Reboot app, in that order.